EXPERIMENT NO-1
LINEAR WAVE SHAPING CIRCUITS -RC Integrator/Low pass Filter Circuits Differentiator/ High pass filter Circuits

Aim

- To observe the response of the designed Integrator/low pass RC circuit and Differentiator/ High pass filter Circuit for the given square waveform for $T_P \ll RC, T_P = RC$ and $T_P \gg RC$ .Where $T_P$is the On time of pulse wave form.
- To observe the frequency response of Integrator/low pass RC circuit and Differentiator/ High pass filter Circuit.

Apparatus Required

Sl No	Name of the Component/Equipment	Specifications
1	Resistor	1KΩ,10KΩ,100KΩ
2	Capacitor	.01µF
3	Function generator	
4	CRO	
5	Bread Board and Connecting wires	

THEORY:

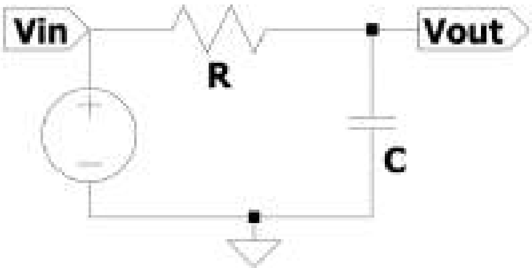
Linear wave shaping circuits are those that use linear components such as resistors, capacitors, and inductors to shape non-sinusoidal input wave forms. RC circuits are used in these linear wave shaping circuits. The two most commonly used circuits adjust the output voltage based on changes in the input voltage, both in terms of its differential and integral.

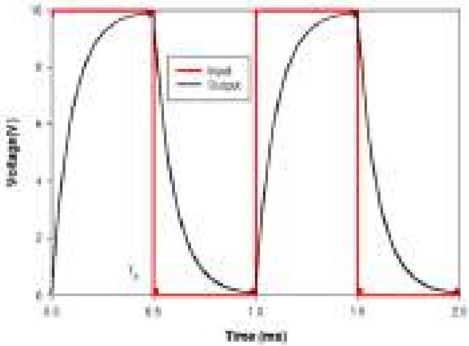
A resistor and capacitor connected in series, with the output voltage applied across the capacitor, make up an RC integrator circuit. The capacitor integrates the new voltage change with its original voltage while it is charging. The term integrator circuit is coined from this. Additionally, the integrator circuit functions as a low pass filter. The study using sinusoidal voltage proves that the reactance of the capacitor decreases with increasing frequency, effectively passing its higher frequency to ground. Low pass circuits should have a short time constant, while integrator circuits typically have long time constants.

A RC differentiator circuit is a series circuit comprising of a resistor, capacitor, and a voltage source, where the output is obtained across the resistor. The voltage across the resistor is commonly referred to as the differential output. The output obtained across the resistor is typically directly proportional to the gradient of the input voltage. Hence, the name differentiator. The differentiator is also referred to as a high-pass RC filter. A high-pass filter selectively allows high frequencies to pass through while attenuating low frequencies, as its name suggests. A short time constant RC circuit refers to a differentiator circuit.

Theory
RC integrator circuit

The circuit diagram is shown in figure.



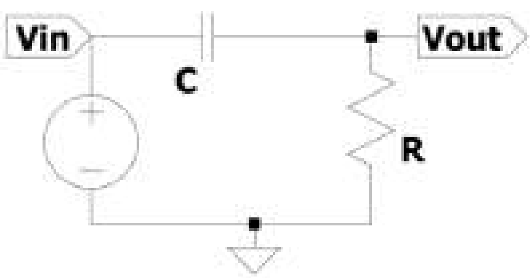


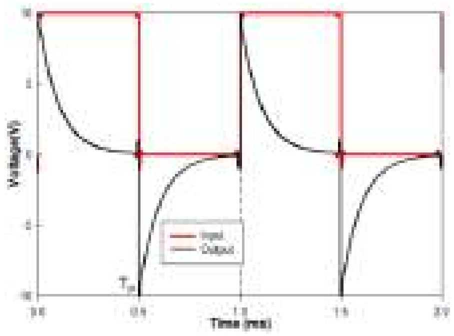
At $t = 0$ time an input pulse voltage of V volt is applied. The capacitor voltage V_C given by $V_C = V - (V \pm V_I)exp^{\frac{-t}{RC}}$.

At Tp the capacitor is charge toV_F, and the input source voltage is zero at that instance Afterwards, the capacitor discharges to a voltage$V_I = (V - V_I)exp^{\frac{-t}{RC}}$.

RC differentiator circuit

Refer to the figure.





During $t < 0 < Tp$, the capacitor attempts to charge from zero to a specific voltage while the input voltage V is applied. As a result, the voltage across the resistor V_R can be determined by subtracting the voltage across the capacitor from the input voltage, following the relationship $V_R = (V_I)exp^{\frac{-t}{RC}}$. At Tp, the input voltage becomes zero and the left plate of the capacitor, which is positive, is abruptly connected to ground. This functions as a negative voltage, and the capacitor attempts to discharge through the resistor to a lower negative voltage. The input voltage V is restored in the subsequent time period.

Procedure

1. Connect the circuit as shown in figures
2. Apply the Square wave input to circuits (Vi = 10 V_{PP}, f =1KHz)
3. Observe the output waveform for (a) $T_P \ll RC, T_P = RC$ and $T_P \gg RC$
4. Verify the values with theoretical calculations
5. Observe the frequency response for $T_P \gg RC$ Low-pass
6. Observe the frequency response for $T_P \ll RC$ High -pass

Calculation

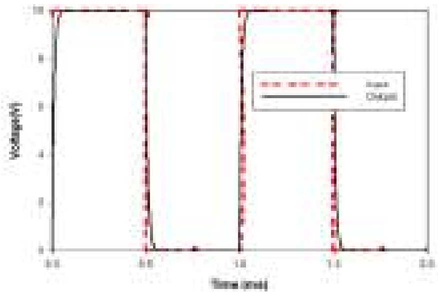
Integrator

Calculation	Theoretical	Practical
$T_P \gg RC$		
$T_P = RC$		
$T_P \ll RC$		
$f_C = \frac{1}{2\pi RC}$		
Differentiator		
Calculation	Theoretical	Practical
$T_P \gg RC$		
$T_P = RC$		
$T_P \ll RC$		
$f_C = \frac{1}{2\pi RC}$-Low pass		
$f_C = \frac{1}{2\pi RC}$-High pass		

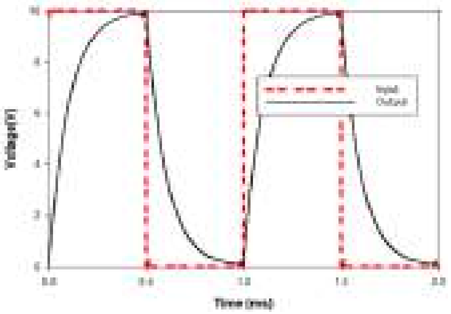
Observation

Simulated Graph-Integrator

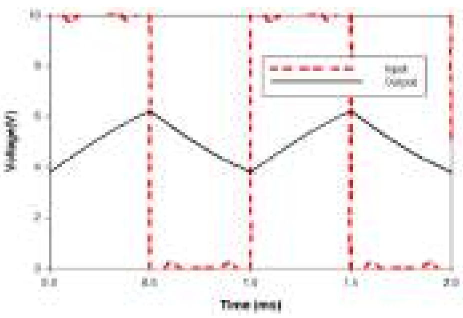
Practical Graph



$RC \ll T_P$



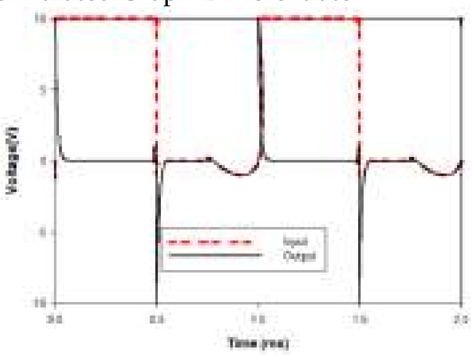
$RC = 2T_P$



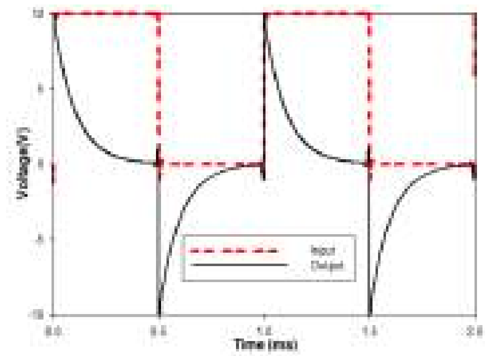
$RC \gg T_P$

Simulated Graph-Differentiator

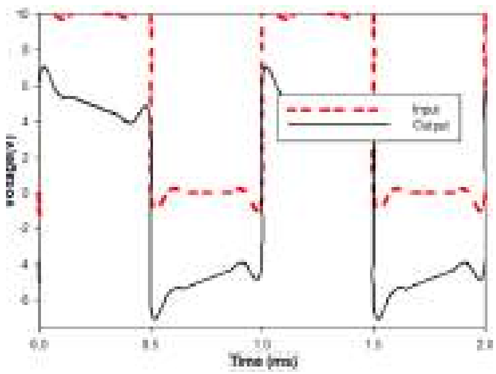
Practical Graph



$RC \ll T_P$



$RC = 2T_P$

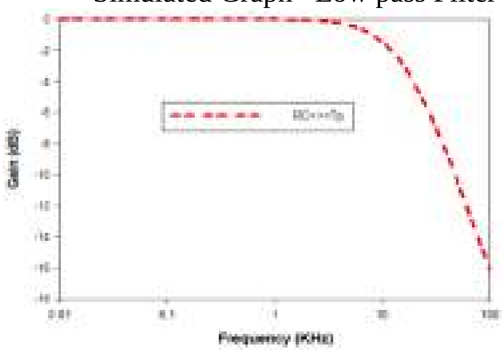


$RC \gg T_P$

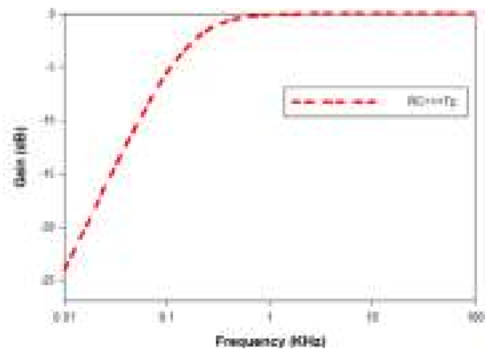
Frequency response

Simulated Graph –Low pass Filter

Practical Graph



Simulated Graph –High Pass Filter



Result:

A RC wave shaping circuit as low pass circuit ,an integrator a high pass circuit and differentiator circuit have been designed and implemented. The response at various time constants and the frequency response have been observed and studied.

EXPERIMENT NO-2A
NON LINEAR WAVE SHAPING CIRCUITS –Clipping Circuits

Aim

- Design and setup diverse clipping circuits using diodes to analyze and visualize the resulting output waveform and transfer characteristics.

Apparatus Required

Sl No	**Name of the Component/Equipment**	**Specifications**
1	Resistor	
2	Diodes	1N4001
3	Function generator	
4	CRO	
5	Bread Board and Connecting wires	

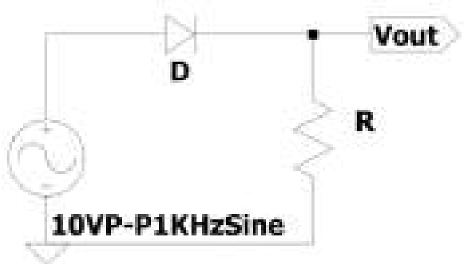
THEORY:
Clipping circuits make use of the diode's property as a switching device. Clipping circuits are a widely used type of non-linear wave shaping circuits. They have use in removing either the positive or negative parts of an input waveform. Additionally, it can be utilised to segment an input waveform based on specific voltage thresholds. By employing a positive clipper, it is possible to generate a square waveform of moderate quality from a sine wave. There are two types of diode clippers: series clippers and shunt clippers. A diode connected in series with the input in a clipper results in what is known as a series clipper. When the diodes are connected in parallel with the input, that clipper is referred to as a shunt clipper. A resistor is employed to regulate the flow of current through the diode. The expression provides the value of the series resistance used in the clipping circuits.

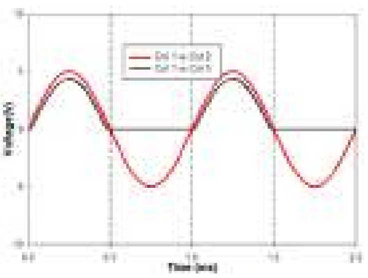
$$R = \sqrt{R_f \times R_r}.$$

Where R_f forward resistance of the diode and R_r reverse resistance of the diode.

Series-Diode Clipper Circuit

The circuit diagram is shown in figure.

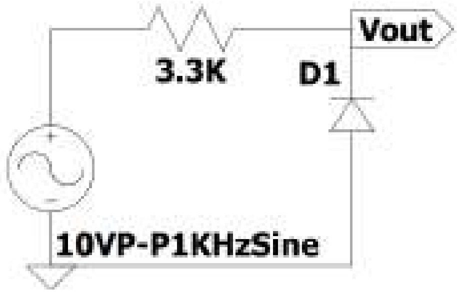


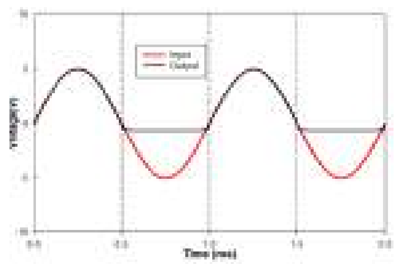


When the diode is forward biased, current flows through the resistor. The diode is forward biased during the positive half cycle, causing the voltage to appear across the resistor. The output is measured across the resistor.

Shunt-Diode Clipper Circuit

Refer to the figure.





In this circuit, the output is measured across the diode and it is connected in parallel. When the diode is off only, there exist a voltage at the output.

Procedure

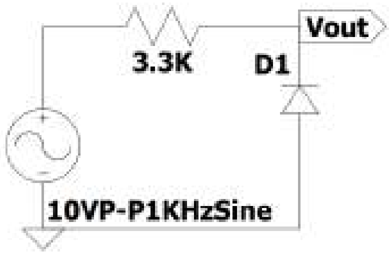
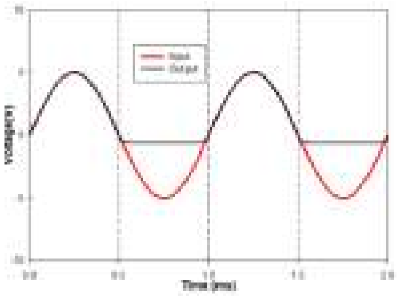
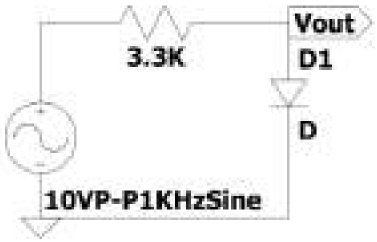
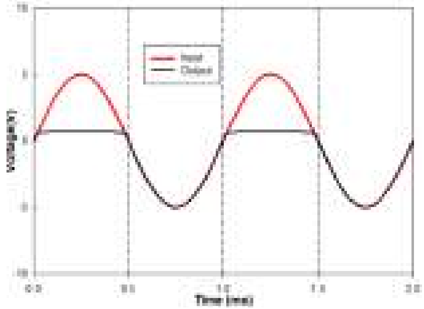
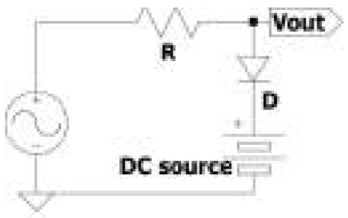
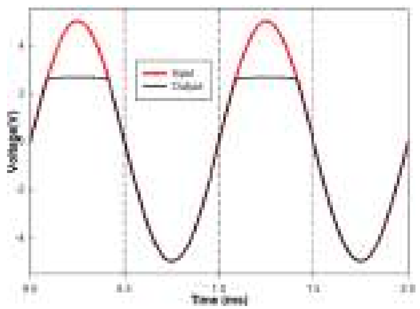
1. As indicated by the circuit diagram, set up the circuit.
2. Use the signal generator to supply the circuit with an input sine wave of 10Vpp and 1 KHz.
3. Examine the CRO's output wave form. Observe the waveforms simultaneously by applying the input to channel X and the output to channel Y. Change the AC-DC coupling to the DC mode.
4. Hold down the XY mode switch and look at the output to see the transfer characteristics.
5. Sketch the result while accounting for the diode drop.

Design

The resistance employed for current limitation in the circuit. Typical values of forward resistance R_f = 30 Ω and of R_r = 300 KΩ, R=3.3 K Ω

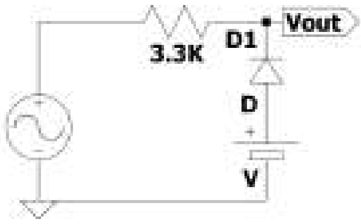
Result:

Output waveforms and transfer characteristics of various clipping circuits are analyzed and plotted.

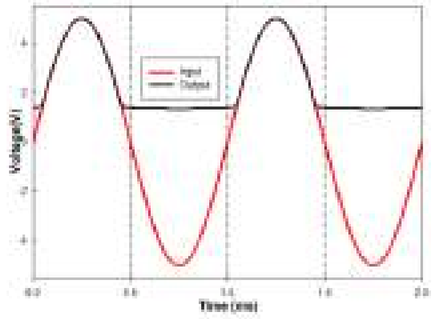
Circuit	Description	Simulated Graph	Practical Graph	Transfer characteristics
Positive clipper with a 0.6V clipping level: 	Only negative-going half cycles of the input are transmitted to the output. The diode effectively bypasses the complete positive half. As a result of the voltage drop across the diode, clipping occurs precisely at +0.6V.			
Negative clipper with a 0.6V clipping level 	Only Positive going half cycles of the input are transmitted to the output. The diode effectively bypasses the complete negative half. As a result of the voltage drop across the diode, clipping occurs precisely at -0.6V.			
Positive clipper with clipping level at +2.6V 	The diode remains in reverse bias until the input voltage exceeds +2V, at which point the input will be reflected at the output. The diode becomes forward biased and the cell voltage is displayed at the output when the input voltage crosses +2V. As the diode is connected in series with the			

cell, the clipping level in practice is +2.6V.

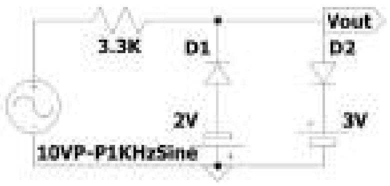
Negative clipper with clipping level at +1.4V



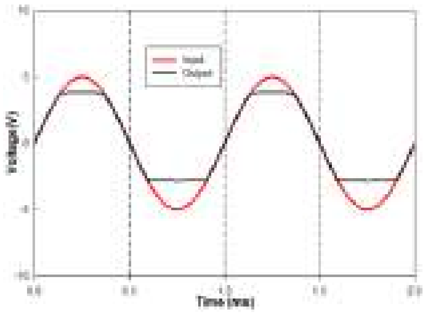
When the input exceeds +1.4V during the positive cycle, the diode is reverse biased and the input is reflected at the output. The diode remains forward biased until the input voltage exceeds +1.4V, at which point the cell voltage is displayed at the output.



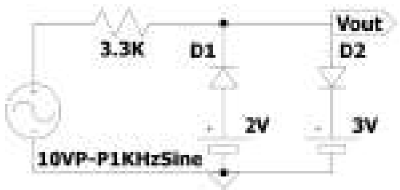
Double clipper with clipping level at +3.6V & -2.6V



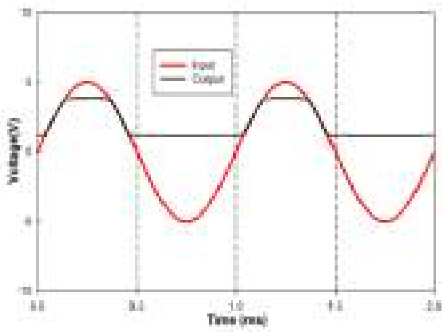
The integration of positive and negative clippers constitutes this circuit. During the positive half cycle of the input, one branch will remain open while the other is rendered active, and conversely. The actual levels of clipping were +3.6V and -2.6V.



Positive slicer with slicing level at +1.4V & +3.6V



During the negative half cycle of the input, diode D1 conducts while diode D2 becomes reverse biased. Therefore, the output remains at +2V. During the positive half cycle of the input, when the input surpasses +2V, D1 becomes reverse biased and the input is reflected at the output. If the output exceeds +3V,



diode D2 will conduct and keep the output at +3V.

EXPERIMENT NO-2A
NON LINEAR WAVE SHAPING CIRCUITS –Clamping Circuits

Aim

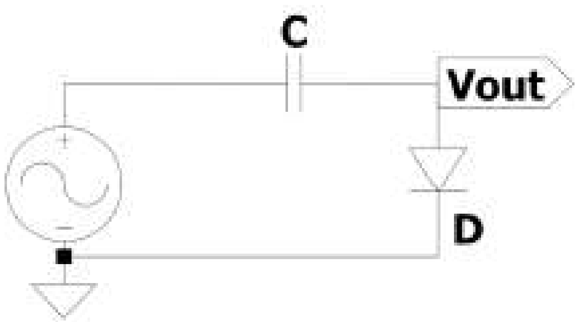
- Design and setup diverse clamping circuits using diodes to analyze and visualize the resulting output waveform and transfer characteristics

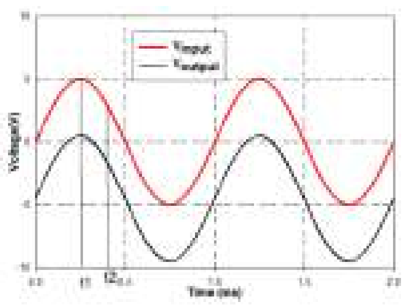
Apparatus Required

Sl No	**Name of the Component/Equipment**	**Specifications**
1	Resistor	
2	Capacitor	
3	Diodes	1N4001
4	Function generator	
5	CRO	
6	Bread Board and Connecting wires	

THEORY:
The term "clamper circuit" derives from the fact that it clamps an AC voltage signal to a DC reference level. In particular, it establishes a reference level of dc voltage around which the absolute minimum or maximum of an AC signal oscillates. Such a system is referred to as a DC restorer or DC inserter circuit. AC signals may take the form of sine waves or other pulse voltage patterns. Ideally, the circuit should modify the DC reference voltage without altering the shape of the AC voltage signal. The voltage source is connected in series with a charged capacitor, which functions as the voltage source, in a clamper circuit. At any given instant, the output voltage is calculated as the algebraic sum of these two voltages. Biasing the diode permits the establishment of clamping at any voltage level. The term for this type of clamping circuit is biased clipper

Negative Clamper Circuit





The output voltage of a negative-voltage clamper circuit is depicted in the figure. The input at time $t1$ is $+5\ V$ relative to ground. As an ideal switch, diode D is biassed forward by this input. In this case, capacitor C charges instantly to 5 V through the very low resistance of the ON diode. The input at time $t2$ is +3 V. The bias voltage of diode $t2$ is calculated as the algebraic sum of the input voltage and the voltage across the capacitor. This voltage reverse biases the diode, causing it to function as an open switch. The input is -5V with respect to ground at time $t3$. There is a difference between the input voltage and the voltage across the capacitor. As a consequence, the output is -10 V.

Procedure

1. As indicated by the circuit diagram, set up the circuit.

2. Use the signal generator to supply the circuit with an input sine wave of 10Vpp and 1 KHz.

3. Examine the CRO's output wave form. Observe the waveforms simultaneously by applying the input to channel X and the output to channel Y. Change the AC-DC coupling to the DC mode.

4. Hold down the XY mode switch and look at the output to see the transfer characteristics.

5. Sketch the result while accounting for the diode drop.

Design

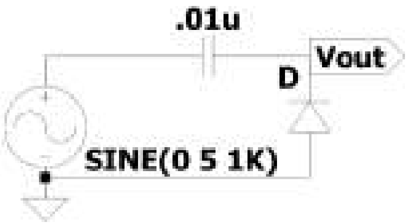
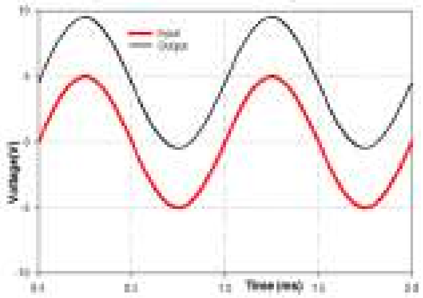
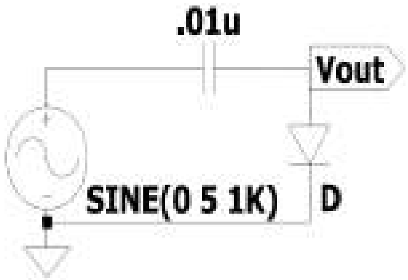
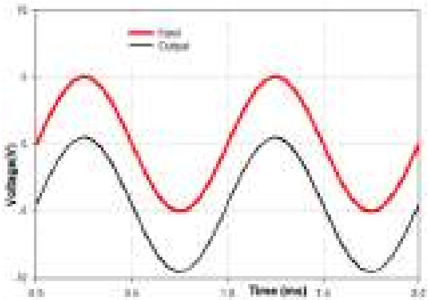
While the diode is in the forward biased state, an adequate amount of time elapses to allow the capacitor to charge to the applied voltage. During this interval, the effective circuit is reduced to a short-time constant differentiator circuit. The capacitor is charged via the source resistor Rs and the diode Rf forward ON resistance. Considering the practical situation, allow the capacitor to attain its peak voltage within a span of five time constants $T_p = 5\tau_c$

Where T_p is half the time period and is the $\tau_c = RC$ time constant,

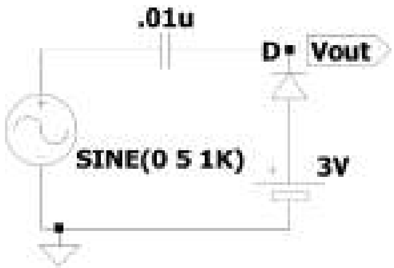
$R = R_s + R_f$,the maximum value of C is calculated from time constant

Result:

Output waveforms and transfer characteristics of various clamping circuits are analyzed and plotted.

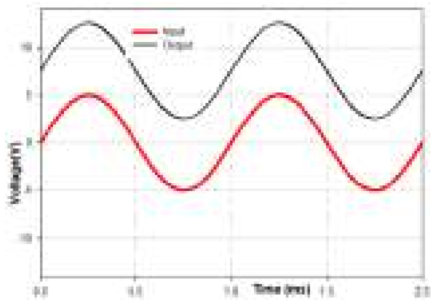
Circuit	Description	Simulated Graph	Practical Graph	Transfer characteristics
Positive clamper with clamping level at 0V 	During the negative half cycle of the input sine wave, the diode conducts, and the capacitor charges to V_m with positive polarity on the right side. The capacitor cannot discharge during the input sine wave's positive half cycle because the diode does not conduct. Thus, the capacitor serves as a dc source of V_m connected in series with the input signal source. The output voltage can then be expressed as:$V_0 = V_m + V_m \sin \omega t$			
Negative clamper with clamping level at 0V 	During a positive half cycle of the input sine wave, the diode conducts and the capacitor charges to $V_m$with a negative polarity on the right side. The capacitor cannot discharge during the negative half-cycle of the input sine wave because the diode does not conduct. The capacitor acts as a $-V_m$ dc source when connected in series with the input .The output voltage can then be expressed as $V_0 = -V_m + V_m \sin \omega t$			

Positive clamper with clamping level at +3V

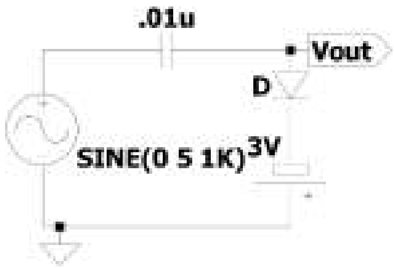


During one negative half cycle of the input sine wave, the capacitor charges through the dc source and diode to $(V_m + 3)$ volts, with the capacitor having positive polarity on the right side. The presence of the dc source limits the capacitor's charging to $(V_m + 3)$ .Output voltage is given by

$V_0 = (V_m + 3) + V_m \sin \omega t$

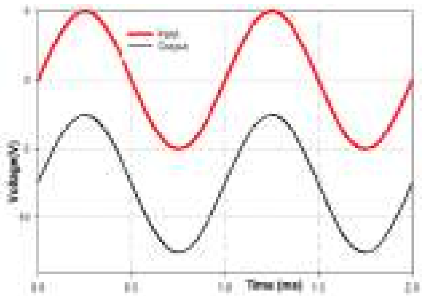


Negative clamper with clamping level at +3V

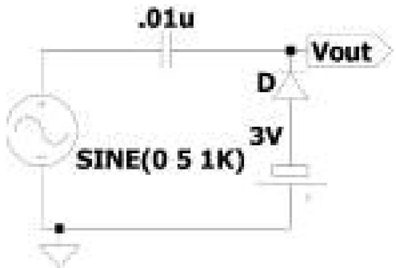


During one positive half cycle of the input sine wave, capacitor charges through the dc source and diode till $V_m + 3)$ volts with negative polarity of the capacitor at the right side. The charging of the capacitor is limited to $V_m + 3)$ volts lts due to the presence of the dc source.

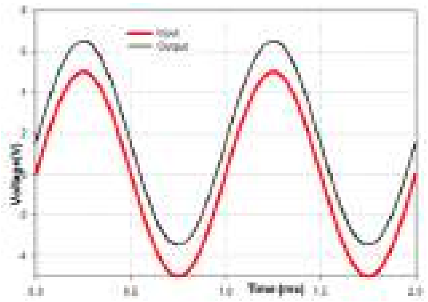
$V_0 = -(V_m + 3) + V_m \sin \omega t$



Positive clamper with clampinglevel at -3V



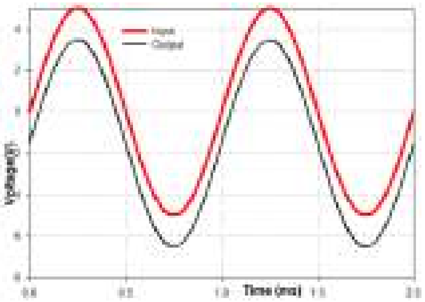
During one negative half cycle of the input sine wave, the capacitor charges through the diode to $(V_m - 3)$ volts with positive polarity at the right. Due to the dc source, capacitor charging is limited to $(V_m - 3)$ volts. The output is $V_0 = (V_m - 3) + V_m \sin \omega t$



Negative clamper with clampinglevel at -3V

At a positive half cycle of the input sine wave, the capacitor charges through diode until it reaches a voltage of $(V_m - 3)$ volts. The polarity of the capacitor is negative on the right side. The capacitor's charging is restricted to $-(V_m - 3)$ volts.

$$V_0 = -(V_m - 3) + V_m \sin \omega t$$



EXPERIMENT NO-3
RC COUPLED COMMON EMITTER AMPLIFIER

Aim

- Design, set-up , and analyse the transient and frequency response of an RC-coupled CE amplifier using NPN transistor.

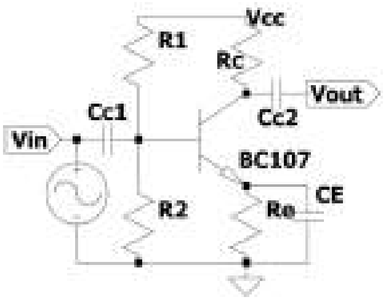
Apparatus Required

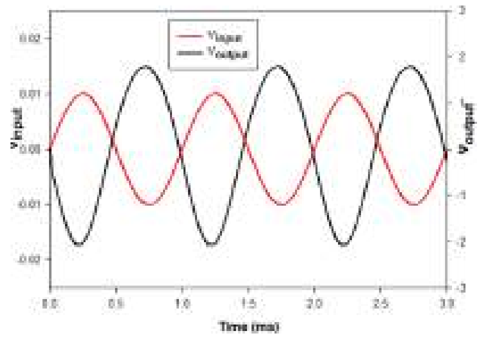
Sl No	Name of the Component/Equipment	Specifications
1	Resistors	
2	Capacitors	
3	Transistor (BJT)	BC107
3	Function generator	
4	CRO	
5	Power supply	
6	Bread Board and Connecting wires	

THEORY:

The RC-coupled CE amplifier is widely used in audio frequency applications in radio and TV receivers. It renders current, voltage, and power amplification. The collector current of a common emitter amplifier is controlled by the base current. Within the circuit diagram, an NPN transistor is configured as a common emitter amplifier. R_1 and R_2 are used for the voltage divider bias of the transistor. Using voltage divider bias ensures stable operation regardless of changes in β using single power supply. The input signal V_{in} is connected to the base through C_{C1}, while the output voltage is connected from the collector through the capacitor C_{C2}. It is important that V_{in} is in the order of V_t (25mv), the thermal equivalent voltage at room temperature. The transistor is chosen based on the frequency of operation and power requirements. Ensure the transistor selected has a minimum guaranteed β or h_{fe} that is greater than or equal to the required voltage gain A_V. The V_{CC} is chosen to be 20% higher than the necessary voltage swing. For instance, if the desired output swing is 10 V, V_{CC} is set at 12 V. You can choose the nominal value of I_{CQ} from the data sheet corresponding to the specifiedh_{fe}.

The circuit diagram is shown in figure.





A slight rise in base current leads to a significant rise in collector current. For the amplifier to work correctly, the emitter-base junction needs to be forward biased, while the collector-base junction should be reverse biased. R_1 and R_2 are chosen so that $\frac{R_B}{1+\beta} \ll R_E \text{ where } R_B = R_1 \parallel R_2$. The Voltage gain $A_V = \frac{V_{out}}{V_{in}} = \frac{-\beta R_C \| R_L}{(1+\beta)r_e}$ where $r_e = \frac{V_t}{I_{CQ}}$

Procedure

1. As indicated by the circuit diagram, set up the circuit and check DC bias conditions. For checking the DC bias conditions, remove the input signal and capacitors from the circuit.

2. Use the signal generator to supply the circuit with an input sine wave of 20mVpp and 1 KHz.

3. Examine the CRO's output wave form. Observe the waveforms simultaneously by applying the input to channel X and the output to channel Y. Change the AC-DC coupling to the DC mode.

4. Maintain a constant input voltage 20mVpp (Sinusoidal) while adjusting the frequency of the input signal from 10Hz to 10 MHz or the highest frequency provided by the generator. Record the amplitude of the output for various frequencies and input the data into a table.

5. Take out the emitter bypass capacitor CE from the circuit and redo steps 3 to 4 with input voltage 20mVpp

6. Sketch the results

Design
Given $V_{CC} = 10V, I_{CQ} = 1mA$ and $\beta = 100$.Also Assume $I_{CQ} = I_{EQ}$

Circuit parameters	Design Equation	Value
R_C	$R_1 = \frac{0.4V_{CC}}{I_{CQ}}$	

R_E	$R_1 = \frac{0.1V_{CC}}{I_{CQ}}$	
R_2	$R_2 = \frac{0.1V_{CC} + V_{BE}}{9I_{BQ}}$	
R_1	$R_1 = \frac{V_{CC} - (0.1V_{CC} + V_{BE})}{10I_{BQ}}$	
C_{C1}	$X_{C1} \leq \frac{R_i + R_S}{10}$	**2uF**
C_{C2}	$X_{C2} \leq \frac{R_C + R_L}{10}$	**2uF**
C_E	$X_E \leq \frac{R_E \parallel (r_e + \frac{R_i \parallel R_S}{1+\beta})}{10}$	**100uF**

TABULAR COLUMN

DC Condition

DC CONDITIONS	V_{RC}	V_{RE}	V_{CE}	V_{R1}	V_{R2}
Theoretical					
Practical					

Frequency Response

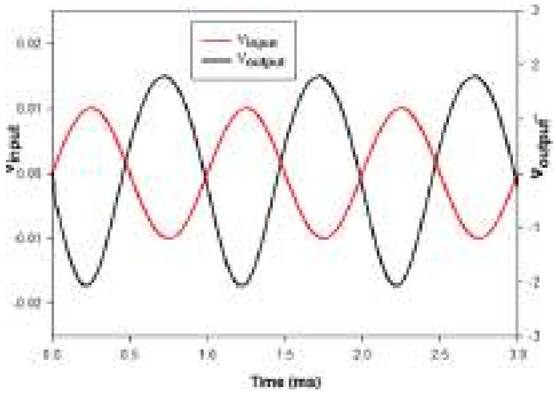
F(Hz) *(10 Frequency points)*	V_{out}	$Log\ F$	$\text{Gain(dB)} = 20\log A_V$

F(Hz) *(10 Frequency points)*	V_{out}	$Log\ F$	$\text{Gain(dB)} = 20\log A_V$

Transient Response Plot

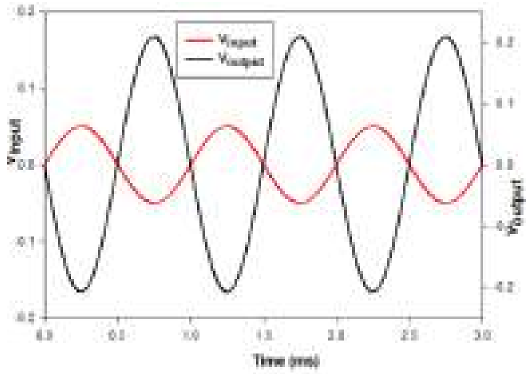
Simulated Graph-With CE

Practical Graph- With CE



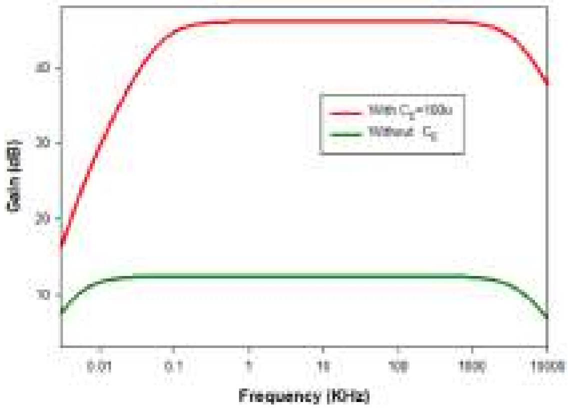
Simulated Graph-Without CE

Practical Graph- Without CE



Frequency Response Plot

Simulated Graph



Practical Graph

Result:

Designed and set up an RC coupled amplifier and studied its frequency response.

RC coupled amplifier with C_E

Mid-band gain of the amplifier =

Bandwidth of the amplifier =

RC coupled amplifier without C_E

Mid-band gain of the amplifier =
Bandwidth of the amplifier =

EXPERIMENT NO-4
CASCADE AMPLIFIER

Aim

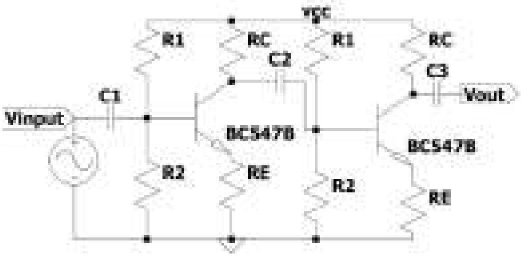
- Design, set-up , and analyse the transient response of a cascaded RC coupled CE amplifier using NPN transistors.

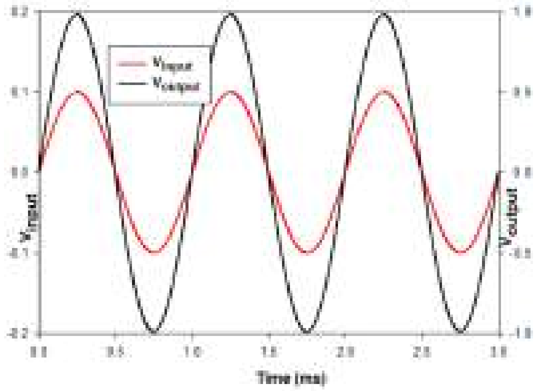
Apparatus Required

Sl No	**Name of the Component/Equipment**	**Specifications**
1	Resistors	
2	Capacitors	
3	Transistor (BJT)	BC107
3	Function generator	
4	CRO	
5	Power supply	
6	Bread Board and Connecting wires	

THEORY:
A cascade refers to a type of multistage amplifier where multiple single stage amplifiers are connected in a series. Often, achieving the desired amplifier performance is challenging with a single stage amplifier due to the limitations of transistor parameters. In these situations, multiple amplifier stages are connected in a series to ensure that the input and output stages meet impedance matching requirements while providing some amplification. The majority of the amplification is then handled by the middle stages. These amplifier circuits are commonly used in the design of microphones and loudspeakers. The design of a single stage RC coupled amplifier using voltage divider bias has already been completed. It is recommended to use two similar stages without an emitter bypassing capacitor in this case. The overall gain of the amplifier is determined as $A_V \leq A_{V1} \times A_{V2}$ where the individual gains of each stage, denoted as A_{V1} and A_{V2}
.

The circuit diagram is shown in figure.





The design of a single stage is replicated in this instance. Even a small increase in base current results in a substantial increase in collector current. In order for the amplifier to function properly, it is necessary to ensure that the emitter-base junction is forward biassed, while the collector-base junction is reverse biased. R_1 and R_2 are chosen so that $\frac{R_B}{1+\beta} \ll R_E$ $where\ R_B = R_1 \parallel R_2$. The Voltage gain $A_{V1} = \frac{V_{out}}{V_{in}} = \frac{-\beta R_C \| R_L}{(1+\beta)r_e + R_E} \cong \frac{R_C}{R_E}$ where $r_e = \frac{V_t}{I_{CQ}}$

Procedure

1. As indicated by the circuit diagram, set up the circuit and check DC bias conditions. For checking the DC bias conditions, remove the input signal and capacitors from the circuit.

2. Use the signal generator to supply the circuit with an input sine wave of 20mVpp and 1 KHz.

3. Examine the CRO's output wave form. Observe the waveforms simultaneously by applying the input to channel X and the output to channel Y. Change the AC-DC coupling to the DC mode.

4. Sketch the results

Design for single stage
Given $V_{CC} = 10V$, $I_{CQ} = 1mA$ and $\beta = 100$.Also Assume $I_{CQ} = I_{EQ}$

Circuit parameters	**Design Equation**	**Value**
R_C	$R_C = \frac{0.4V_{CC}}{I_{CQ}}$	
R_E	$R_E = \frac{0.1V_{CC}}{I_{CQ}}$	

R_2	$R_2 = \frac{0.1V_{CC} + V_{BE}}{9I_{BQ}}$	
R_1	$R_1 = \frac{V_{CC} - (0.1V_{CC} + V_{BE})}{10I_{BQ}}$	
C_{C1}	$X_{C1} \leq \frac{R_i + R_S}{10}$	**2uF**
C_{C2}	$X_{C2} \leq \frac{R_{out1} + R_{Inp2}}{10}$	**2uF**
C_{C3}	$X_{C2} \leq \frac{R_{C2} + R_L}{10}$	**2uF**

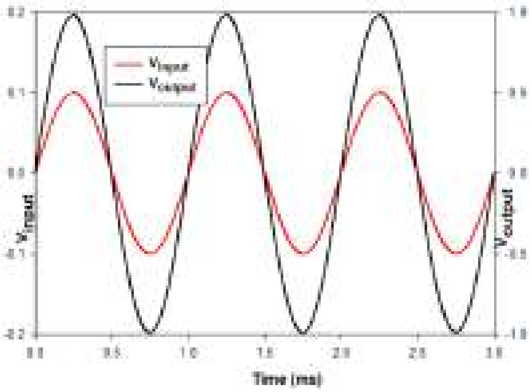
TABULAR COLUMN

DC Condition

DC CONDITIONS	V_{RC}	V_{RE}	V_{CE}	V_{R1}	V_{R2}
Theoretical					
Practical					

Transient Response Plot

Simulated Graph



Practical Graph

Result:

Designed and set up an RC coupled amplifier and studied its frequency response.

Mid-band gain of the amplifier =

EXPERIMENT NO-5
CASCODE AMPLIFIER

Aim

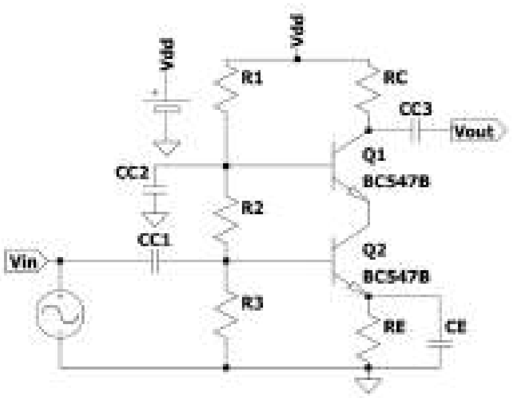
- Design, set-up, and analyse the transient response of a Cascode amplifier using NPN transistors.

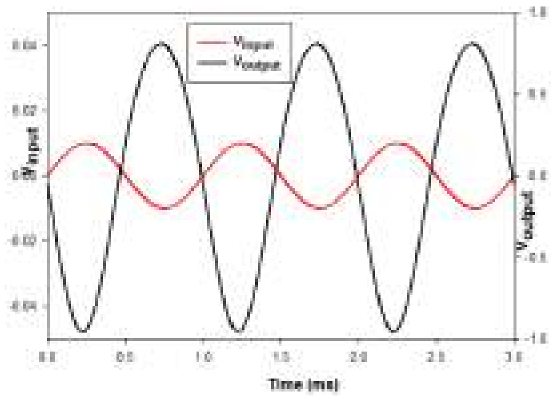
Apparatus Required

Sl No	Name of the Component/Equipment	Specifications
1	Resistors	
2	Capacitors	
3	Transistors(BJT)	BC107
3	Function generator	
4	CRO	
5	Power supply	
6	Bread Board and Connecting wires	

THEORY:
A cascade amplifier consists of a common emitter amplifier and a common base amplifier stages in cascade. The main advantage of this circuit is its low internal capacitance, which limits the gain at high frequencies. The Cascode amplifier has the ability to amplify a wider range of frequencies compared to the CE amplifier. The Voltage gain $A_V = \frac{V_{out}}{V_{in}} = -g_m R_C \parallel R_L$ where $g_m = \frac{I_{CQ}}{V_t}$

The circuit diagram is shown in figure.





Procedure

1. As indicated by the circuit diagram, set up the circuit and check DC bias conditions. For checking the DC bias conditions, remove the input signal and capacitors from the circuit.

2. Use the signal generator to supply the circuit with an input sine wave of 20mVpp and 1 KHz.

3. Examine the CRO's output wave form. Observe the waveforms simultaneously by applying the input to channel X and the output to channel Y. Change the AC-DC coupling to the DC mode.

4. Sketch the results

Design for single stage

Given $V_{CC} = 12V, I_{CQ} = 2mA$ and $\beta = 100$.Also Assume $I_{CQ} = I_{EQ},, V_{CE1} = V_{CE2} = 0.35 \times V_{CC}$

Circuit parameters	**Design Equation**	**Value**
R_C	$R_C = \frac{0.2V_{CC}}{I_{CQ}}$	
R_E	$R_E = \frac{0.1V_{CC}}{I_{CQ}}$	
R_3	$R_3 = \frac{0.1V_{CC}+V_{BE2}}{8I_{BQ}}$ $0.1V_{CC} + V_{BE2} = V_{R3}$	
R_1	$R_1 = \frac{V_{CC} - (0.1V_{CC} + V_{CE2} + V_{BE1})}{10I_{BQ}}$ $V_{CC} - (0.1V_{CC} + V_{CE2} + V_{BE1}) = V_{R1}$	
R_2	$R_2 = \frac{V_{CC} - (V_{R1} + V_{R3})}{9I_{BQ}}$	
C_{C1}	$X_{C1} \leq \frac{R_{i1} + R_S}{10}$	**2uF**
C_{C2}	$X_{C2} \leq \frac{R_{InpCB2}}{10}$	**2uF**
C_{C3}	$X_{C3} \leq \frac{R_{C2} + R_L}{10}$	**2uF**
C_E	$X_E \leq \frac{R_E \parallel (r_{e1} + \frac{R_{i1} \parallel R_S}{1+\beta})}{10}$	**100uF**

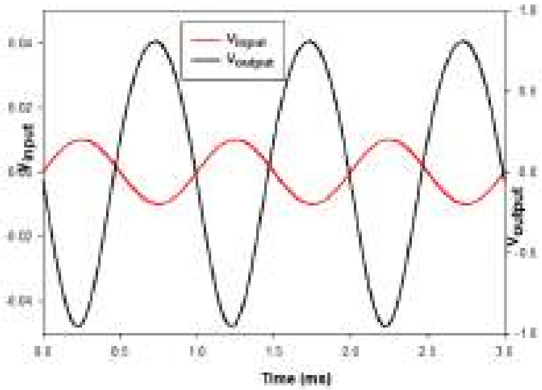
TABULAR COLUMN

DC Condition

DC CONDITIONS	V_{RC}	V_{RE}	V_{CE1}	V_{CE2}	V_{R2}	V_{R2}	V_{R3}
Theoretical							
Practical							

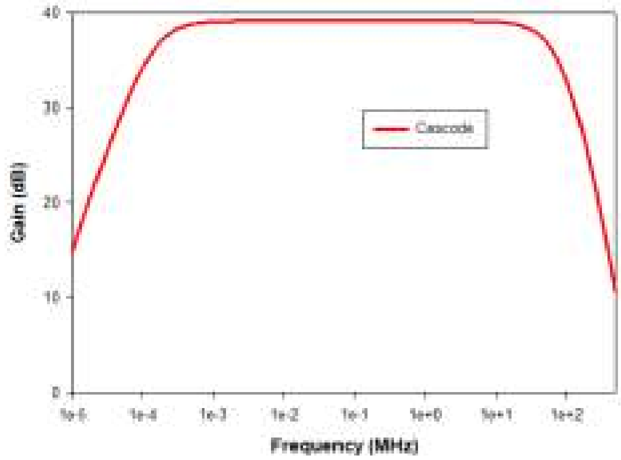
Transient Response Plot

Simulated Graph



Practical Graph

Simulated Frequency Response Plot



Result:

Designed and set up an Cascode Amplifier and studied its transient response.

Mid-band gain of the amplifier =

EXPERIMENT NO-6
RC PHASE SHIFT OSCILLATOR

Aim

- Design, set-up, and analyse the output of a RC phase shift oscillator using NPN transistors.

Apparatus Required

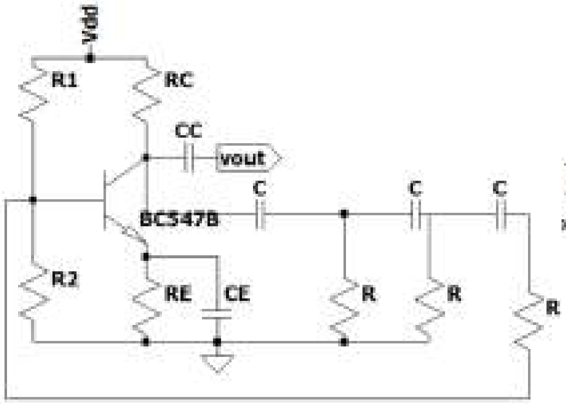
Sl No	**Name of the Component/Equipment**	**Specifications**
1	Resistors	
2	Capacitors	
3	Transistors(BJT)	BC107
3	Function generator	
4	CRO	
5	Power supply	
6	Bread Board and Connecting wires	

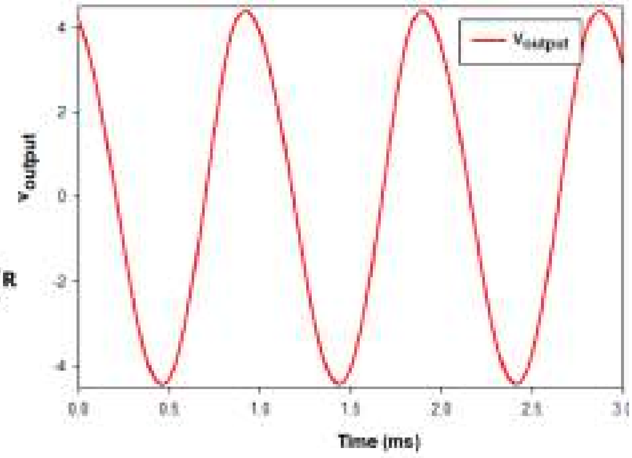
THEORY:

An oscillator is an electronic circuit that requires only a DC supply as input in order to generate an AC signal voltage. The circuit components determine the frequency of the generated signal. An amplifier, a frequency selective network, and positive feedback from the output to the input are necessary for an oscillator. Given amplifier's gain (A) and feedback factor (β) then $A\beta = 1$, is the Barkhausen criterion for sustained oscillation. The signal is in phase when the gain is at unity. The voltages at the base and collector of a common emitter amplifier have a 180^0phase shift when used with a resistive collector load. The feedback network that connects the base and collector needs to add an extra 180^0phase shift at a specific frequency.

In the given diagram, three sections of phase shift networks are utilised to ensure that each section imparts an approximate 60^0 degrees of phase shift at the resonant frequency. Through careful analysis, the resonant frequency f can be mathematically expressed by the following equation

$$f = \frac{1}{2\pi CR\sqrt{6 + \frac{4R_C}{R}}}$$





The three section RC network provides a β value of 1/29. Therefore, the amplifier's gain should be 29.

Procedure

1. As indicated by the circuit diagram, set up the circuit and check DC bias conditions. For checking the DC bias conditions, remove the input signal and capacitors from the circuit.

3. Examine the CRO's output wave form. Change the AC-DC coupling to the DC mode.

4. Sketch the results

Design for $f = 1KHz$
Given $V_{CC} = 12V, I_{CQ} = 2mA$ and $\beta = 100$.Also Assume $I_{CQ} = I_{EQ},, V_{CE} = 0.4 \times V_{CC}$

Circuit parameters	Design Equation	Value
R_C	$R_C = \frac{0.4V_{CC}}{I_{CQ}}$	
R_E	$R_E = \frac{0.1V_{CC}}{I_{CQ}}$	
R_2	$R_2 = \frac{0.1V_{CC}+V_{BE}}{9I_{BQ}}$	
R_1	$R_2 = \frac{V_{CC} - V_{R2}}{10I_{BQ}}$	

C_{C2}	$X_{C2} \leq \frac{R_{InpCB2}}{10}$	**2uF**
C_E	$X_E \leq \frac{R_E \parallel (r_{e1} + \frac{R_{i1} \parallel R_S}{1+\beta})}{10}$	**10u or less**
R	R is taken as $2R_C$.	
C	$C = \frac{1}{2\pi f R\sqrt{6 + \frac{4R_C}{R}}}$	

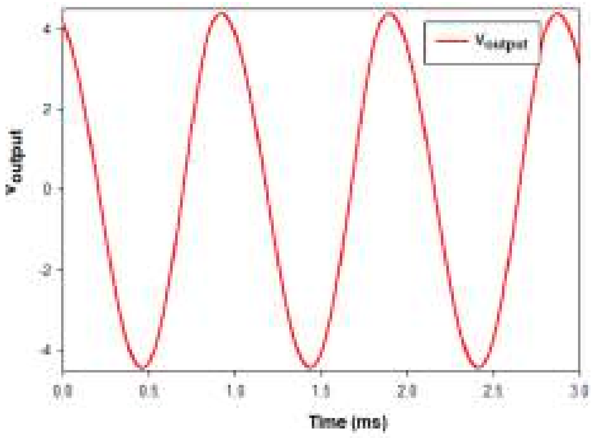
TABULAR COLUMN

DC Condition

DC CONDITIONS	V_{RC}	V_{RE}	V_{CE1}	V_{CE2}	V_{R2}	V_{R2}	V_{R3}
Theoretical							
Practical							

Output Plot

Simulated Graph



Practical Graph

Result:

Designed and set up an CRC Phaseshift Oscillator and studied its output response.

Measured frequency of the oscillator =